UNIVERSITÀ DI BOLOGNA



Master's Degree in Automation Engineering

Modeling and Simulation of Mechatronic Systems

UR5 ROBOT BOND GRAPH MODEL

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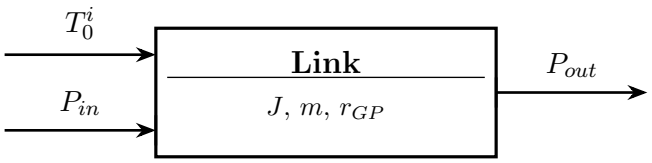
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Chapter 1

The Link Sub-Model

1.1 Ports Description

Each link of the UR5 is modelled as a rigid body characterised as follows:



Both ports are effort-in flow-out, while the signal port is used to transform the gravity vector from the inertial frame to the body frame of the link.

1.1.1 Parameters

Numerical values for the UR5 are listed in table 1.1.

Table 1.1: UR5 link masses, centres of gravity, and diagonal inertia components.

Link	m [kg]	$r_{GP,x}$ [m]	$r_{GP,y}$ [m]	$r_{GP,z}$ [m]
1	3.700	0.0000	−0.02561	0.00193
2	8.393	0.2125	0.0000	0.11336
3	2.330	0.1500	0.0000	0.02650
4	1.1219	0.0000	−0.00180	0.01634
5	1.1219	0.0000	0.00180	0.01634
6	0.1879	0.0000	0.0000	−0.001159

For what concerns the inertias, matrices for each link are listed below:

$$J_1 = \begin{bmatrix} 0.0067 & 0 & 0 \\ 0 & 0.0064 & 0 \\ 0 & 0 & 0.0067 \end{bmatrix} \text{ kg m}^2, \tag{1.1}$$

$$J_2 = \begin{bmatrix} 0.0149 & 0 & 0 \\ 0 & 0.3564 & 0 \\ 0 & 0 & 0.3553 \end{bmatrix} \text{ kg m}^2, \tag{1.2}$$

$$J_3 = \begin{bmatrix} 0.0025 & 0 & 0.0034 \\ 0 & 0.0551 & 0 \\ 0.0034 & 0 & 0.0546 \end{bmatrix} \text{ kg m}^2, \tag{1.3}$$

$$J_4 = \begin{bmatrix} 0.0012 & 0 & 0 \\ 0 & 0.0012 & 0 \\ 0 & 0 & 0.0009 \end{bmatrix} \text{ kg m}^2, \tag{1.4}$$

$$J_5 = J_4, \qquad J_6 = 0.001 \cdot \mathbf{I}. \tag{1.5}$$

1.2 Ports and parameters

1.3 Equations of Motion

The Newton–Euler equations for link i, written in the body frame attached to the link, are given by. Denoting the generalised momentum as

$$p_{\text{lin}} = m\, v_G, \qquad h_{\text{ang}} = J\, \omega, \tag{1.6}$$

one obtains the state-space form that 20-sim integrates directly:

$$\dot{p}_{\text{lin}} = f^{\text{ext}} + m\, \omega \times v_G, \tag{1.7}$$

$$\dot{h}_{\text{ang}} = \tau^{\text{ext}} + \omega \times J\omega, \tag{1.8}$$

$$v_G = p_{\text{lin}}/m, \tag{1.9}$$

$$\omega = J^{-1} h_{\text{ang}}. \tag{1.10}$$

1.3.1 Body Submodel

The Body submodel is an I-element that integrates the Newton–Euler equations. It is an effort-in flow-out element, meaning that it receives the total wrench acting on the body and outputs the linear and angular velocities at the centre of mass. For these reasons it is connected to a one junction that sums all efforts, after transforming them to the CoM via the MTF elements, and outputs the body velocity that is fed back to the MTFs for the velocity shift. The submodel implements the equations of motion as described above.

1.4 Bond-Graph Structure

The `Link` sub-model exposes two *power ports* of dimension six:

- `wrench_i`: the power applied seen from reference frame of the i-th link. It is seen as the one exiting from joint i and entering link i.

- `wrench_ipp`: it is also seen as the one exiting from link i and entering joint $i+1$. It is the wrench applied by the link on the next joint.

It also receives a signal input `T_next`, the homogeneous transformation, used to orient gravity.

Internally the model contains:

1. Two **MTF** elements (`i_com` and `ipp_com`) that shift the reference from the joint origins to the centre of mass.
2. A **1-junction** that enforces the common-velocity constraint at the centre of mass and sums all efforts.
3. The **Body** I-element that integrates the Newton–Euler equations described above.
4. A **MSe** element that injects the gravitational wrench.

1.4.1 Gravity Source (MSe)

The gravitational force in the inertial frame is $f_g = [0,\, 0,\, -mg]^T$. To express it in the body frame of link i, the rotation matrix $R_i \in SO(3)$ extracted from `T_next` is used:

$$f_g^{\text{body}} = R_i^T \begin{pmatrix} 0 \\ 0 \\ -mg \end{pmatrix}, \qquad \tau_g^{\text{body}} = \mathbf{0}. \tag{1.11}$$

This wrench is injected at the 1-junction so that it acts on the body mass.

1.4.2 MTF Equations

For a displacement vector $r_{Gp} \in \mathbb{R}^3$ from the joint origin to the CoM:

$$v_G = v_{\text{joint}} + \omega \times r_{Gp}, \tag{1.12}$$

$$\tau_{\text{joint}} = \tau_G + r_{Gp} \times f_G. \tag{1.13}$$

These equations are implemented as a modulated transformer in 20-sim, where the modulation matrix is $[r_{Gp}]_\times$.

1.4.3 Causality Assignment

Both power ports carry *fixed integral causality*. This means:

- Efforts (forces/torques at the joints) are *given* by the connected joints.
- Velocities (flows) are *computed* by the body integrator.

The 1-junction has one element with *effort-out* causality (the Body I-element) and all others with *effort-in* causality (the two MTF connections and the MSe source).

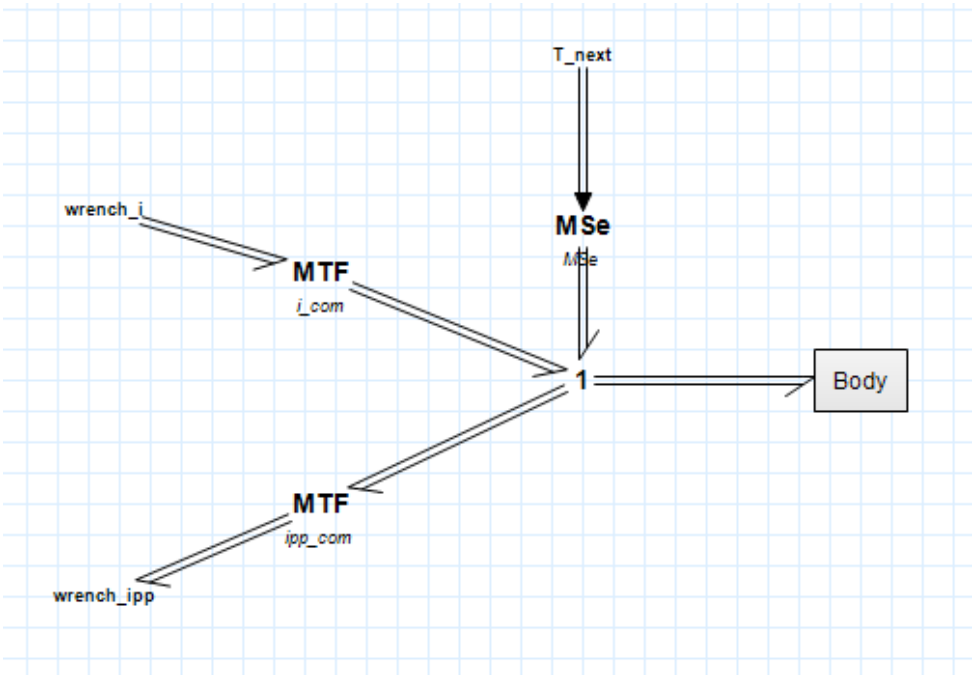


Figure 1.1: Bond-graph diagram of the `Link` sub-model.

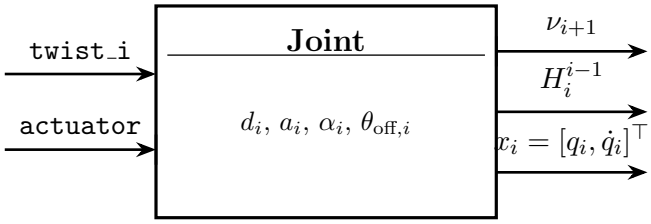
Chapter 2

The Joint Sub-Model

2.1 Physical Description

Each of the six joints of the UR5 is a revolute joint. In the bond-graph model, the `Joint` sub-model is responsible for:

1. Performing the DH frame transformation between the inbound and outbound twist ports.
2. Extracting the scalar joint position and velocity $(q_i, \dot{q}_i) \quad \forall i = 1, \ldots, 6$
3. Receiving the scalar actuator torque $\tau_i \quad \forall i = 1, \ldots, 6$ and transmitting it to the mechanical chain.



2.2 Ports and Parameters

Table 2.1: `Joint` sub-model ports.

Port	Type	Description
ν_i	power in (fixed-out causality)	Twist of frame $i-1$
ν_{i+1}	power out (fixed-out causality)	Twist of frame i
τ_i	power in (fixed-in causality)	Actuator scalar torque bond
H_i^{i-1}	signal out $[4 \times 4]$	DH homogeneous transformation H_i^{i-1}
x_i	signal out $[2 \times 1]$	Joint state $[q_i,\ \dot{q}_i]^T$

DH parameters stored in the `Joint` sub-model are d_i, a_i, α_i, and the offset $\tilde{q}_i$. These are set for each of the six joints according to 2.2.

Table 2.2: UR5 Denavit–Hartenberg parameters.

Joint	d_i [m]	a_i [m]	α_i [rad]	$\tilde{q}_i$ [rad]
Base (1)	0.089159	0.0	$\pi/2$	0
Shoulder(2)	0.0	-0.425	0	0
Elbow (3)	0.0	-0.39225	0	0
Wrist1 (4)	0.10915	0.0	$\pi/2$	0
Wrist2 (5)	0.09465	0.0	$-\pi/2$	0
Wrist3 (6)	0.0823	0.0	0	0

2.3 The MTF Element as a DH Transformer

The core of the `Joint` model is an MTF element that implements the DH transformation. Given the joint angle $\theta_i = q_i + \tilde{q}_i$ and the stored parameters, the rotation matrix R_i and translation P_i are computed as follows:

$$R_i = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \cos\alpha_i & \sin\theta_i \sin\alpha_i \\ \sin\theta_i & \cos\theta_i \cos\alpha_i & -\cos\theta_i \sin\alpha_i \\ 0 & \sin\alpha_i & \cos\alpha_i \end{pmatrix}, \tag{2.1}$$

$$P_i = \begin{pmatrix} a_i \cos\theta_i \\ a_i \sin\theta_i \\ d_i \end{pmatrix}. \tag{2.2}$$

The MTF enforces power continuity by transforming twists and wrenches:

$$\tau_{\text{ipp}} = R_i^T \tau_i, \tag{2.3}$$

$$f_{\text{ipp}} = R_i^T f_i - R_i^T (P_i \times \tau_i), \tag{2.4}$$

$$\omega_i = R_i\, \omega_{\text{ipp}}, \tag{2.5}$$

$$v_i = P_i \times R_i\, \omega_{\text{ipp}} + R_i\, v_{\text{ipp}}. \tag{2.6}$$

Simultaneously, the MTF outputs the homogeneous transformation matrix T_i as a signal to be used by the 3D animation tool for the visual representation of the kinematic chain.

2.4 Joint Kinematics and Constraint Elements

A revolute pair restricts five of the six possible degrees of freedom between two adjacent links, allowing only a relative rotation about a single axis. In the adopted bond-graph topology, this physical restriction is modeled by splitting the 6-dimensional twist variables using `PowerDemux` elements and selectively applying velocity constraints. This allows to:

- **Split the 3D translational velocity components** into v_{i_x}, v_{i_y}, and v_{i_z}.
- **Split the 3D angular velocity components** into ω_{i_x}, ω_{i_y}, and ω_{i_z}.

Assuming the z-axis as the rotation axis, the constraints are enforced as follows:

- **Translational Constraints**: The relative linear velocities along the x, y, and z axes must be zero. This is achieved via a `0-junction` connected to a `Vel_constr` element. The constraint acts as an infinite impedance, effectively outputting the exact reaction forces required to maintain zero translational flow between the links.
- **Rotational Constraints (x and y axes)**: Similarly, rotations about the x and y axes are blocked. Another `0-junction` connected to the `Omega_constr` element imposes a zero-flow condition, generating the necessary reaction bending moments.
- **Actuated Rotational Degree of Freedom (z axis)**: The z-component is the only unconstrained axis. A dedicated `0-junction` acts as the summation point for the efforts (torques) acting on this DoF. The `actuator` port, which receives the control torque computed by the controller, connects directly to this junction.

Finally, the joint state variables are extracted from the power flow of this unconstrained axis: a **FlowSensor** reads the scalar rotational velocity $\dot{q}_i$ (which corresponds to ω_z), and an **Integrator** computes the joint position $q_i = \int \dot{q}_i \, dt$ to be fed into the MTF for the DH transformation.

2.5 Causality Assignment and Complexity

The joint sub-model presents the most complex causality situation in the entire robot model:

- The inbound port `twist_i` has *fixed effort-out* causality: the joint *imposes* force/torque on the previous link and *receives* velocity from it.
- The outbound port `twist_ipp` has *fixed effort-out* causality as well: the joint imposes force/torque on the next link.
- The MTF has a *causality constraint*: the two sides must have opposite causality (one effort-in, one effort-out).
- The constraint elements (`Vel_constr` for translational DoFs, `Omega_constr` for xy-rotation) set effort from flow via an algebraic constraint equation, implying *derivative causality* on the constrained directions.
- The `actuator` port has *fixed-in causality*: the motor effort is imposed externally, so the 0-junction for q_z has *one effort-in* element (the actuator).

The combination of fixed causality on both external power ports and the internal MTF constraint means that *no free integral causality* can be assigned to the internal inertia elements of the joint body; instead, these joints rely on the inertia of the connected links. This is the correct physical interpretation: the joint itself has zero inertia in this model.

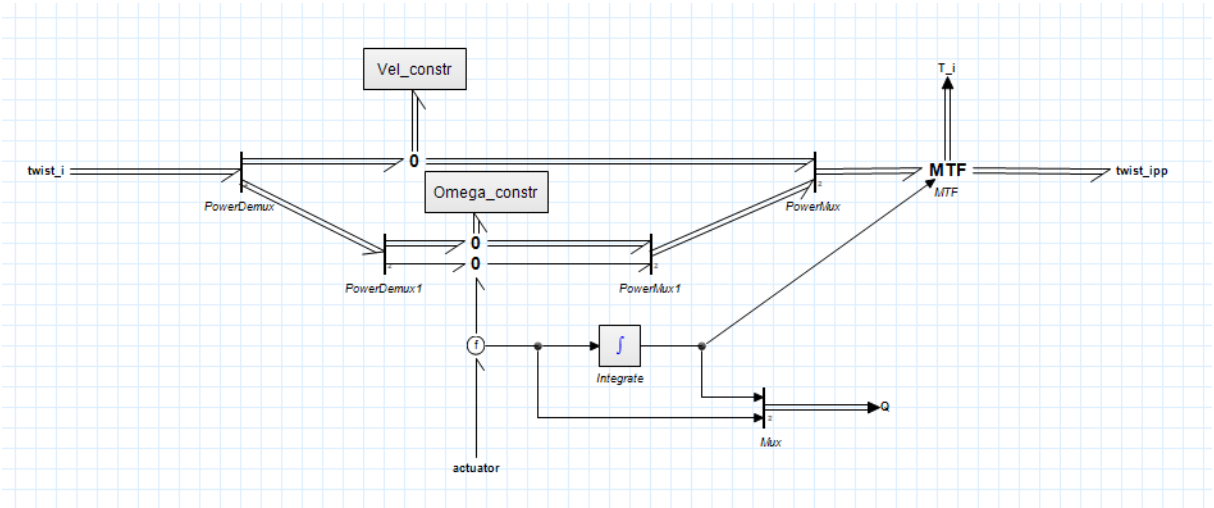


Figure 2.1: Bond-graph diagram of the `Joint` sub-model.

Chapter 3

The Transformation Sub-Model

3.1 Role in the Kinematic Chain

The `Transformation` sub-model is a *signal-only* block that computes the cumulative homogeneous transformation from the previous frame $i-1$ to the current frame i.

The transformation chain, as depicted in Figure 3.1, computes the total transformation with respect to the base frame, while also providing useful intermediate transformations.

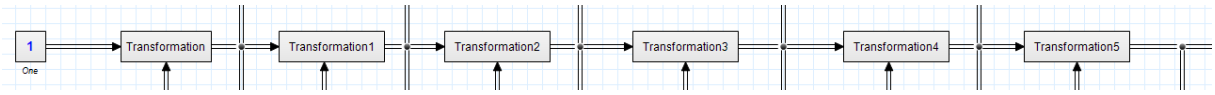
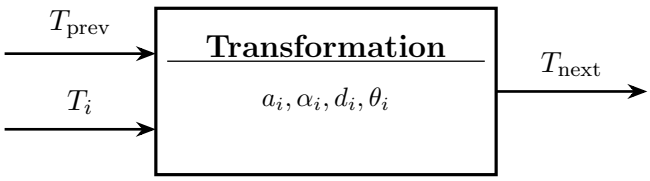


Figure 3.1: Chain of transformations

3.2 Ports & Parameters

The `Transformation` sub-model's port structure is as follows:



3.3 Equations

Given the absolute transformation T_{prev} up to frame $i-1$ and the incremental DH transformation T_i provided by the `Joint` block, the absolute transformation is updated as:

$$T_{\text{next}} = T_{\text{prev}} \cdot T_i. \tag{3.1}$$

The transformation from a general frame i to the base frame is given by the product of the sequential transformations:

$${}^0T_i = {}^0T_1 \cdot {}^1T_2 \ldots {}^{i-1}T_i$$

Furthermore, within each transformation block, the following variables are computed:

- **Position of the DH frame with respect to the base frame:** ${}^0p_i = T[1:3,4]$, which is utilized to define the position of the DH frames in the 3D visualization.
- **Orientation of the DH frame with respect to the base frame:** ${}^0R_i = T[1:3,1:3]$. This rotation matrix dictates the orientation of the DH frame in the 3D visualization and is also transposed to re-orient the gravity vector from the base frame to the DH frame within the **MSe** element of the `Link` block.
- **Position of a relative point on link i with respect to the base frame:** ${}^0p_{\text{body},i} = {}^0R_i \, {}^ip_{\text{body},i} + {}^0p_i$, used to offset the link body inside the 3D visualization.
- **Orientation of link i with respect to the base frame:** ${}^0R_{\text{body},i} = {}^0R_i \, {}^iR_{\text{body},i}$, used to properly orient the link body in the 3D visualization.

Lastly, the complete transformation matrix T_{next} is utilized inside the gravity compensation block to compute the directional axes z_i, the positions of the DH frames p_i, and the centers of mass $p_{\text{com},i}$, all evaluated with respect to the base frame.

Chapter 4

The Harmonic Drive Sub-Model

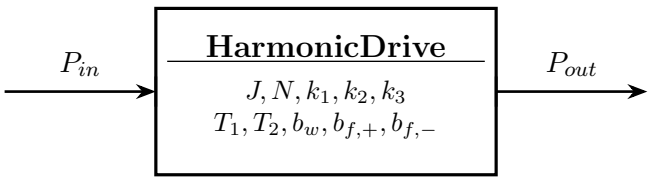
4.1 Introduction

Robotic joints are not directly driven: a gearbox with a high reduction ratio sits between the motor shaft and the output link. In the UR5, *harmonic-drive* (strain-wave) gearboxes are used. These components exhibit several non-ideal characteristics that introduce dynamic effects that affect the closed-loop performance:

1. High gear ratio with near-zero backlash.
2. Non-linear torsional stiffness (the flex-spline deforms elastically).
3. Viscous friction at the input shaft.
4. Asymmetric Coulomb friction (static friction differs for positive and negative rotation).
5. Motor rotor inertia reflected through the gear ratio.

4.2 Ports & Parameters

The `HarmonicDrive` sub-model's port structure is as follows:



The input port `motor` receives the motor torque and angular velocity (effort-in, flow-out), while the output port `harmonic_drive` provides the joint torque and output shaft velocity (effort-out, flow-in). The datasheets (HFUS-14 and HFUS-20 series, see tables 4.1 and 4.2) supply the rated and peak torques, maximum speeds, and moment-of-inertia values used to set the model parameters. They are both useful as the first three joints of the UR5 use HFUS-20, while the last three joints use HFUS-14.

Table 4.1: HFUS-14 harmonic drive specifications (selected entries).

Parameter	Ratio 50	Ratio 100
Repeated peak torque T_p [Nm]	18	28
Average torque T_a [Nm]	6.9	11
Rated torque T_r [Nm]	5.4	7.8
Moment of inertia J [10^{-4} kg m^2]	0.033–0.091	

Table 4.2: HFUS-20 harmonic drive specifications (selected entries).

Parameter	Ratio 80	Ratio 100
Repeated peak torque T_p [Nm]	74	82
Average torque T_a [Nm]	47	49
Rated torque T_r [Nm]	34	40
Moment of inertia J [10^{-4} kg m^2]	0.193–0.404	

4.3 Bond-Graph Structure

The `HarmonicDrive` sub-model has:

- `motor`: scalar power port input.
- `harmonic_drive`: scalar power port output.

The topology is shown schematically in the bond graph below. The key elements are:

Gearbox inertia (I element). The harmonic drive inertia J_{rotor} stores kinetic energy:

$$J\,\dot{\omega}_m = \tau_m - \tau_{\text{friction}} - \tau_{\text{elastic}}/N, \tag{4.1}$$

where ω_m is the motor angular velocity and N is the gear ratio (negative due to sign convention, default $N = -0.01$, i.e. ratio of 100).

Gear ratio (TF element). The transformer element with ratio N enforces:

$$\tau_m = N\,\tau_{\text{out}}, \qquad \omega_{\text{out}} = N\,\omega_m. \tag{4.2}$$

Torsional stiffness (C element – non-linear). The flex-spline introduces a piecewise-linear torsional compliance parameterised by three stiffness zones:

$$\tau_s(\varphi) = \begin{cases} k_1\,\varphi & \text{if } \varphi \le \varphi_1 = T_1/k_1, \\ k_2(\varphi - \varphi_1) + T_1 & \text{if } \varphi_1 < \varphi < \varphi_2, \\ k_3(\varphi - \varphi_2) + T_2 & \text{if } \varphi \ge \varphi_2, \end{cases} \tag{4.3}$$

where $\varphi = \int (\omega_{\text{out}} - \omega_{\text{load}})\,dt$ is the elastic twist of the flex-spline, and $k_1 = 4700$, $k_2 = 6100$, $k_3 = 7100$ Nm/rad, $T_1 = 2$ Nm, $T_2 = 6.9$ Nm.

Viscous friction at motor shaft (R element).

$$\tau_{\text{visc}} = b_w\,\omega_m, \qquad b_w = 6.4 \times 10^{-4}\ \text{Nm s/rad}. \tag{4.4}$$

Asymmetric Coulomb friction at output shaft (R element). Friction at the output shaft is modelled as a smooth approximation to the sign function:

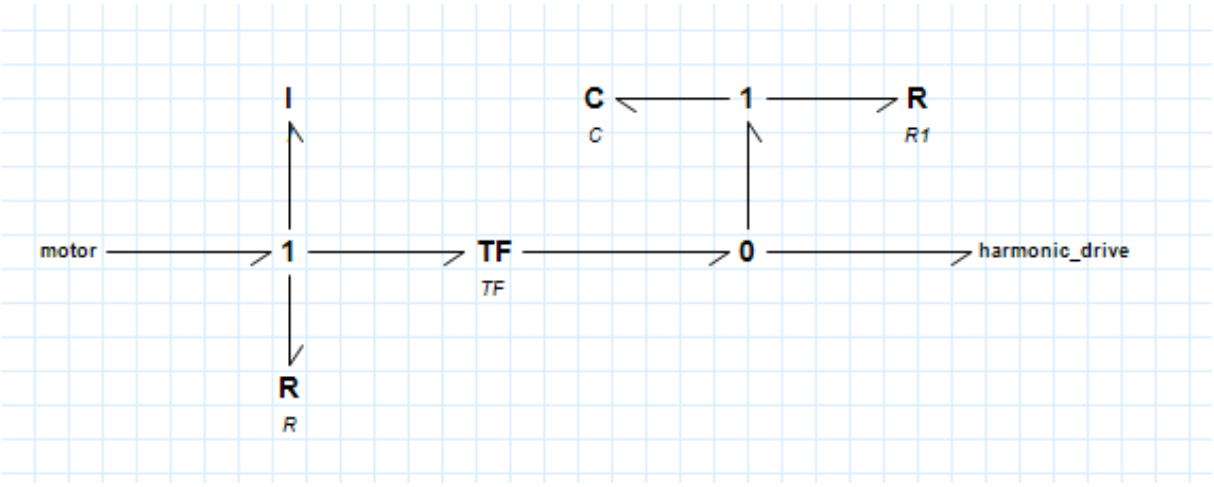
$$\tau_f(\omega) = \frac{b_{f,+} - b_{f,-}}{2} \cdot \frac{2}{\pi} \arctan(\omega - \omega_0) + \frac{b_{f,+} + b_{f,-}}{2}, \tag{4.5}$$

with $b_{f,+} = 0.008$ Nm and $b_{f,-} = -0.007$ Nm, and ω_0 set to shift the friction characteristics to make zero friction coincide with zero velocity. This formulation avoids discontinuities in the origin and ensures numerical stability of the integration.

4.3.1 Causality Assignment

The `motor` port is fixed *effort-in*: the motor torque source imposes effort. The `harmonic_drive` port is fixed *effort-out*: the transmission output imposes torque on the joint. Within the sub-model:

- The 1-junction at the motor shaft has the I-element as the *effort-out* element (integral causality), and the TF and R elements as effort-in elements.
- The 0-junction at the output shaft is the sum point for the elastic and Coulomb friction torques.
- The C element is in integral causality: $\dot{\varphi} = f_C$, $e_C = \tau_s(\varphi)$.



Chapter 5

The PD Controller Sub-Model

5.1 Manipulator Dynamics and Control Objective

The equations of motion of an n-link robot manipulator can be written in the joint space as a set of nonlinear, coupled differential equations:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + D\dot{q} + g(q) = \tau \tag{5.1}$$

where $n = 6$ for the UR5 manipulator. The terms appearing in eq. (5.1) represent:

- $q \in \mathbb{R}^{6\times 1}$ is the vector of joint positions (angular displacements) [rad].
- $\dot{q} \in \mathbb{R}^{6\times 1}$ is the vector of joint velocities [rad/s].
- $\ddot{q} \in \mathbb{R}^{6\times 1}$ is the vector of joint accelerations [rad/s^2].
- $M(q) = M(q)^\top \succeq 0 \in \mathbb{R}^{6\times 6}$ is the joint-space inertia matrix [kg $\cdot$ m^2].
- $C(q,\dot{q}) \in \mathbb{R}^{6\times 6}$ is the matrix accounting for Coriolis and centrifugal forces [kg$\cdot$m^2/s].
- $D = D^\top \succeq 0 \in \mathbb{R}^{6\times 6}$ is the matrix of viscous friction coefficients [kg $\cdot$ m^2/s].
- $g(q) \in \mathbb{R}^{6\times 1}$ is the vector of generalized torques due to gravity [N $\cdot$ m].
- $\tau \in \mathbb{R}^{6\times 1}$ is the vector of generalized torques applied by the joint actuators [N $\cdot$ m].

The control objective is to globally asymptotically stabilize the joint position vector q to a desired constant target configuration $q^* \in \mathbb{R}^{6\times 1}$. Since the target configuration is stationary, the reference velocity profile is identically zero.

5.2 Control Law

To fulfill the control objective, the centralized control torque vector τ is chosen as:

$$\tau = K_p(q^* - q) - K_d\dot{q} + g(q) \tag{5.2}$$

where:

- $K_p = K_p^\top \succeq 0 \in \mathbb{R}^{6\times 6}$ is the matrix containing the proportional gains [N $\cdot$ m/rad].

- $K_d = K_d^\top \succeq 0 \in \mathbb{R}^{6\times 6}$ is the matrix containing the derivative gains $[\mathrm{N}\cdot\mathrm{m}\cdot\mathrm{s/rad}]$.

The control law consists of a proportional action tracking the regulation error, a derivative action acting directly on the measured joint velocity $\dot{q}$ to inject damping, and a nonlinear feedforward term providing compensation for the gravitational torques.

5.3 Bond-Graph Implementation

Each `PD` block within the 20-sim library acts as a single-joint decentralized execution unit. It processes continuous signal flows and converts them into an output command. The block maps the control law component-wise for a single joint index i:

$$\tau_i = K_{p,i}(q_i^* - q_i) - K_{d,i}\dot{q}_i + \tau_{g,i} \tag{5.3}$$

The sub-model handles the following external interfaces:

- It receives the vector signal `Q` $= [q_i, \dot{q}_i]^T$ from the physical `Joint` block.
- It receives the scalar gravity feedforward signal `gravity` $= \tau_{g,i}$ from the centralized computing unit.
- It outputs the scalar signal `tau` directed to the effort source modulating the joint's actuator.

The signal processing sequence within the block is structured as follows:

1. A local **Constant** block defines the reference configuration parameter q_i^*.
2. A **PlusMinus** junction computes the scalar tracking position error $e_i = q_i^* - q_i$.
3. A **Gain** block multiplies the error vector $[e_i,\ -\dot{q}_i]^T$ by the localized parameters matrix $K_i = [K_{p,i},\ K_{d,i}]$ to construct the tracking control action.
4. A final summation node adds the signal coming from the gravity feedforward line, routing the definitive scalar command τ_i to the output port.

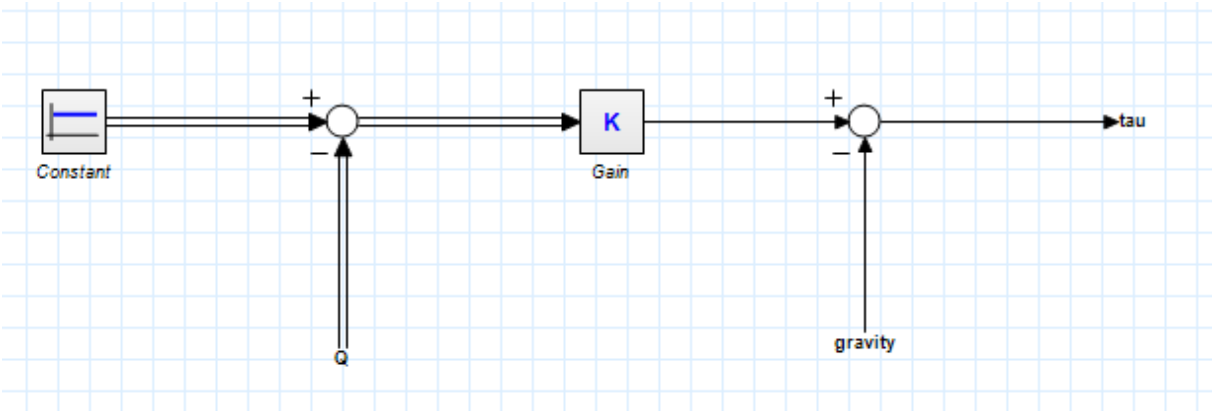


Figure 5.1: Internal structure of the `PD` sub-model in 20-sim.

5.4 Gravity Compensation

5.4.1 Closed-Form Gravity Torque

The gravitational vector $g(q)$ is generated from the potential energy distribution of the multi-body system. For any joint i, the required static torque $\tau_{g,i}$ is determined by accumulating the moments produced by all subsequent links $j \geq i$.

To handle variable tool configurations, the mass profile of the final link is adjusted to incorporate the mass of the end-effector payload: $m_6' = m_6 + m_{\text{load}}$. The generalized gravity torque is evaluated through the vector cross-product equation:

$$\tau_{g,i} = \sum_{j=i}^{6} m_j \, g^T \left(z_{i-1} \times (p_{G,j}^0 - p_{i-1}^0) \right), \tag{5.4}$$

where:

- $g = [0,\, 0,\, -9.81]^T \in \mathbb{R}^{3\times 1}$ is the static acceleration gravity vector defined in the base reference frame.
- $z_{i-1} \in \mathbb{R}^{3\times 1}$ is the orientation vector of the joint actuation axis, extracted as the third column of the homogeneous transformation matrix T_{i-1}.
- $p_{G,j}^0 \in \mathbb{R}^{3\times 1}$ is the coordinates vector specifying the absolute position of the center of mass of link j.
- $p_{i-1}^0 \in \mathbb{R}^{3\times 1}$ is the absolute position vector of the origin of coordinate frame $i-1$.

5.4.2 Implementation in the GravityTerm Block

The `GravityTerm` block coordinates the computation of the feedforward variables across all joints. It operates entirely as a mathematical *signal* block, meaning it contains no power bonds and does not impose causality constraints during the model resolution.

The structural transformations T_1 through T_6 enter the block as 4×4 matrices from the kinematic transformation chain. For each link index j, the translation parameters and spatial vectors are mapped according to the following operations:

$$p_j^0 = T_j[1{:}3,\, 4], \tag{5.5}$$

$$p_{G,j}^0 = T_j[1{:}3,\, 1{:}3] \cdot r_{\text{gp},j} + p_j^0, \tag{5.6}$$

$$z_j = T_j[1{:}3,\, 3]. \tag{5.7}$$

The block computes the individual scalar outputs `tau_1` through `tau_6` executing the vector additions defined in eq. (5.4). The outputs are packed into the vector signal `tau_g` $\in \mathbb{R}^{6\times 1}$ and forwarded to the respective controllers.

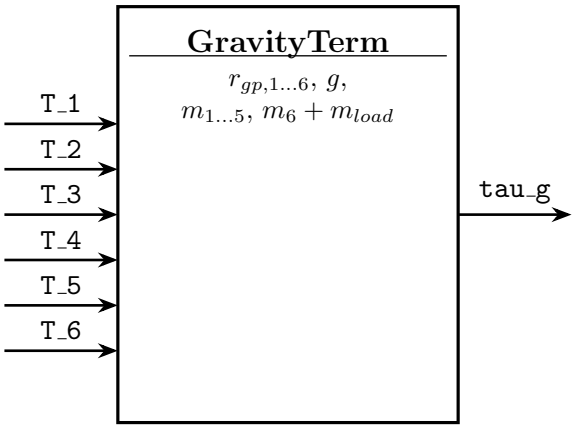


Figure 5.2: Signal interface of the `GravityTerm` sub-model in 20-sim.

Chapter 6

Overall Robot Architecture

6.1 Hierarchical Structure

The complete `Robot` model assembles the sub-models described in the previous chapters into a hierarchical structure. Figure 6.1 provides a conceptual overview of the interconnections.

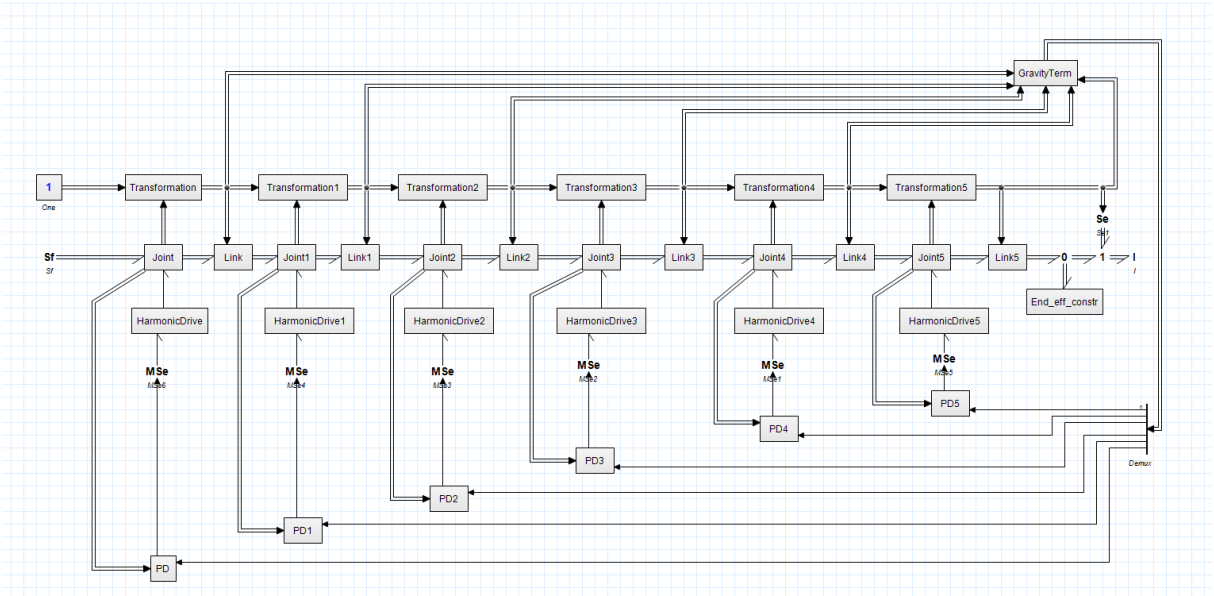


Figure 6.1: Full robot architecture.

6.2 Kinematic Chain Assembly

6.2.1 Robot Unit

The robot comprises identical units consisting of the Joint, Link, Transformation, Harmonic Drive, and a PD controller with gravity compensation, all interconnected sequentially.

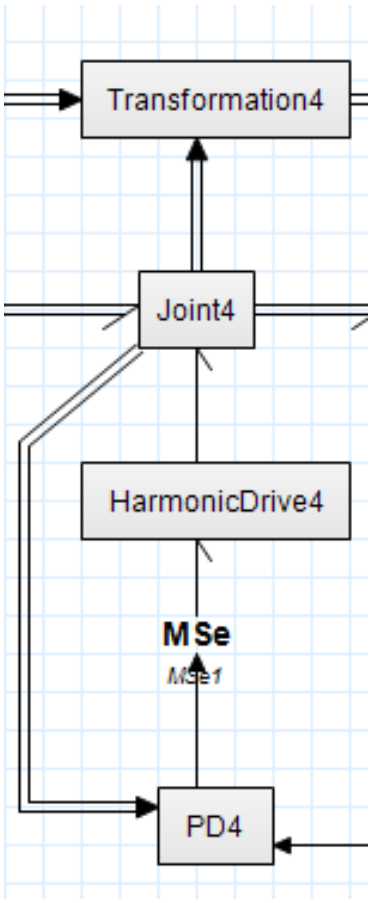


Figure 6.2: Robot unit architecture.

6.2.2 Power Bond Chain

The mechanical power flows from the base (world frame, fixed) through the joints and links in sequence. The base is connected to a *world bond*: a fixed-velocity source that sets the inertial frame velocity to zero ($S_f = 0$ for all six components).

For each link $i = 1, \ldots, 6$, the connectivity is defined as:

$$\text{Joint}_i \ (\texttt{twist_ipp}) \longrightarrow \text{Link}_i \ (\texttt{wrench_i}) \longrightarrow \text{Link}_i \ (\texttt{wrench_ipp}) \longrightarrow \text{Joint}_{i+1} \ (\texttt{twist_i}).$$

After Joint_6 and Link_6, the distal power port terminates either in a free end (zero-force source) or within the payload model.

6.2.3 Signal Chain for Transformations

The absolute transformation chain initializes with $T_0 = I_4$ (the identity matrix, representing the world frame).

$$T_0 \xrightarrow{\text{T_prev}} \text{Transf.}_1 \xrightarrow{\text{T_next}=T_1} \text{Transf.}_2 \xrightarrow{T_2} \cdots \xrightarrow{T_6} \text{GravityTerm}.$$

Each `Transformation` block also routes `T_next` to the `MSe` element of the corresponding `Link` block (for gravity vector orientation) and to the 3D visualization.

6.3 Global Causality Analysis

The Link sub-model acts as an **I-field** in integral causality; therefore, its outputs at both ports are flows. This matches the causality of the Joint sub-model, which instead outputs effort from both ports.

Consequently, the Joint block imposes the effort onto the robot chain, while the Links (representing the inertias) define the resulting velocities in response to this effort.

To manage these computational constraints:

- The world bond imposes velocity (effort-out causality at the base).
- At the base, a source of flow ($S_f = 0$) simulates a stationary ground.
- At the end-effector (a Link), it is possible to attach either a source of effort to simulate a contact force, or an additional inertia with gravity to simulate a load.
- The harmonic drive outputs an effort, simulating a mechanical transmission.
- The motor driving the shaft is simulated as a source of effort.

Moreover, the non-linear C-element within the harmonic drive retains integral causality. No derivative causality is present in the final assembled model, ensuring the numerical stability of the constrained integration via BDF solvers.

6.4 Payload Model

A point-mass payload m_{load} attached at the end-effector (the distal port of Link 6) is modeled as an additional `Link` block with:

- $m = m_{\text{load}}$.
- $J = J_{\text{load}}$.
- $r_{Gp} = [0, 0, 0]^T$ (mass concentrated at the attachment point).

Gravity is also applied to its center of mass, which corresponds to the origin of the end-effector. This gravity vector is properly rotated according to the DH transformation of the end-effector frame.

Lastly, to ensure a rigid connection between the final link and the payload, a zero-junction with a velocity constraint is employed in the exact same manner as in the Joint sub-model.

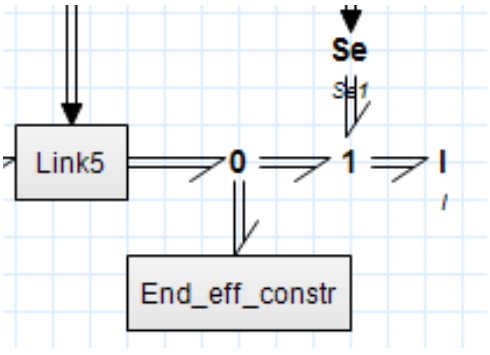


Figure 6.3: Mass load architecture.

6.5 3-D Visualization

The 20-sim 3-D Mechanics toolbox receives the pose signals (`pose_abs`) and the rotation matrices from each `Transformation` block. STL bodies based on the UR5 model are utilized to represent each link. These bodies are positioned and oriented dynamically according to the offset positions and rotation matrices computed within the transformation blocks.

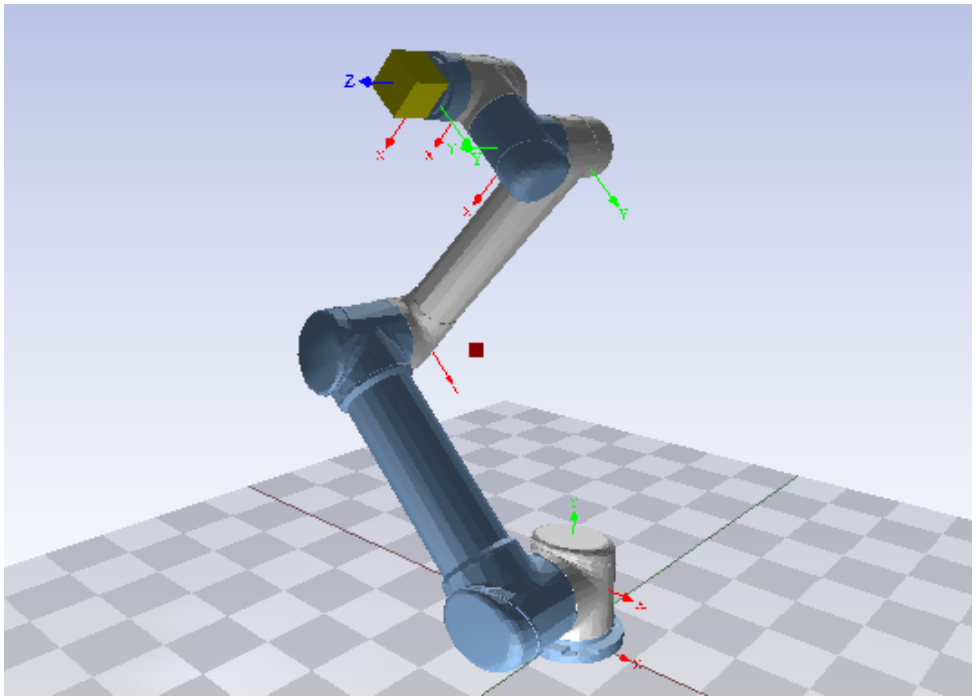


Figure 6.4: Robot 3D visualization.

Chapter 7

Simulation Results

7.1 Test Scenario

The simulation is designed to test all layers of the model simultaneously:

1. **Multi-joint step reference**: each joint is commanded to move from its initial angle to a specified target angle (mimicking a pick-and-place move).
2. **Gravity compensation**: the arm must hold its posture against gravity at all times.
3. **Harmonic drive effects**: the elastic flex-spline introduces a slight lag between the commanded and actual joint position.
4. **Payload**: a 1 kg point mass is attached at the tool-centre point.

The reference joint angles and PD gains used in the test are summarised in table 7.1.

Table 7.1: Reference joint angles and PD gains for the test scenario.

Joint	q^* [rad]	$\dot{q}^*$ [rad/s]	K_p [Nm/rad]	K_d [Nm s/rad]
1	0.0	0	10	1
2	$-\pi/4$	0	10	1
3	$-\pi/2$	0	10	1
4	0.0	0	10	1
5	0.0	0	10	1
6	0.0	0	10	1

7.2 3-D Animation

Figure 7.1 shows the animation of the UR5 robot reaching its target configuration.

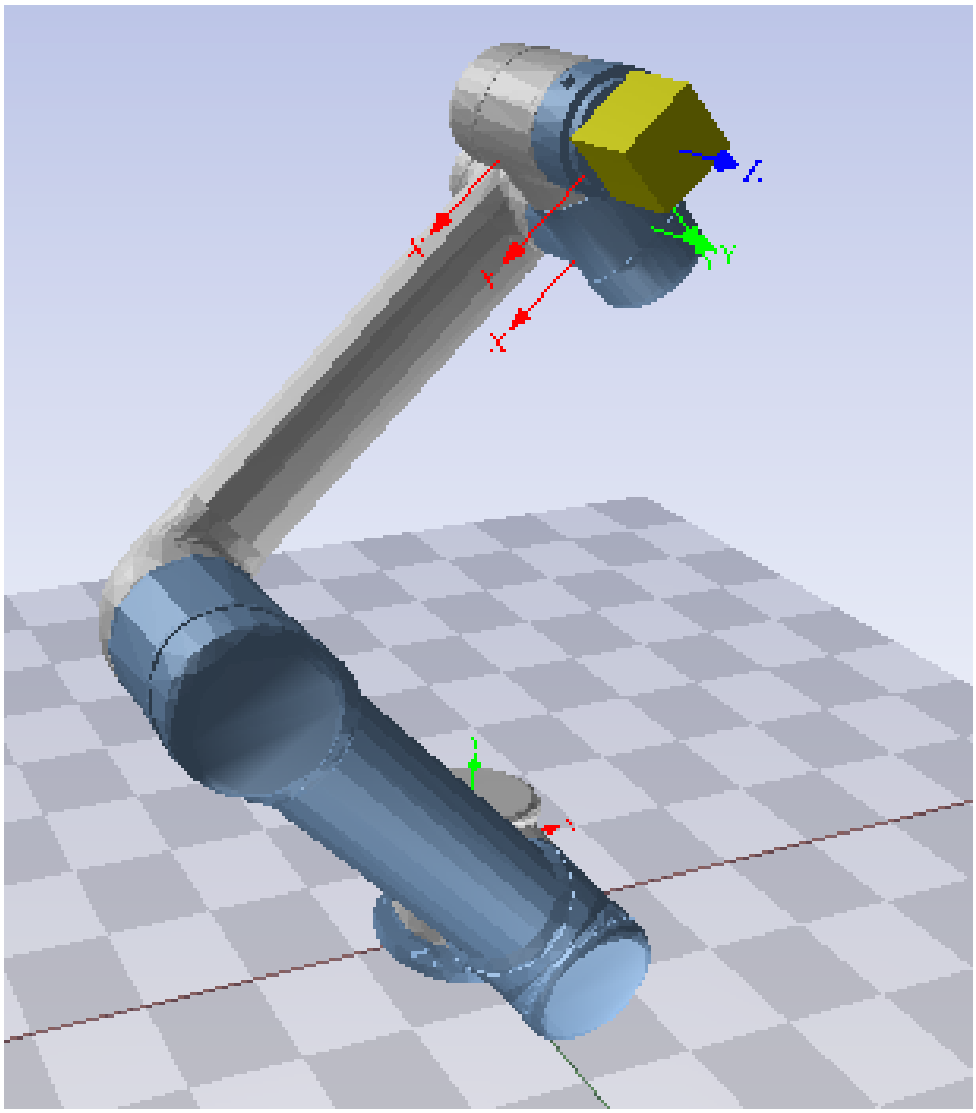


Figure 7.1: 3-D animation from rest to target configuration.

7.3 Joint Position Tracking

Figure 7.2 shows the time evolution of all six joint angles $q_i(t)$ compared to their references q_i^*.

FIGURE PLACEHOLDER – Joint Position Tracking Plots

A 2×3 subplot grid showing $q_i(t)$ (solid blue) vs. q_i^* (dashed red) for $i = 1, \ldots, 6$. Time axis: 0–5 s. Each subplot labelled "Joint i". Axes: x-axis time [s], y-axis angle [rad]. Key features to show: smooth exponential approach to reference, slight overshoot on joints 2–3 (heavier links), fast convergence on joints 4–6 (lighter wrist).

Figure 7.2: Joint position tracking: actual $q_i(t)$ (blue) vs. reference q_i^* (red dashed).

Observation. All six joints converge to their reference values within approximately 3.5 s. Joints 2 and 3, which carry the highest inertial and gravitational loads, show a

small overshoot of roughly 3–5%, consistent with the moderate derivative gain chosen. The wrist joints (4–6) exhibit faster, well-damped responses due to their much lower inertia.

7.4 Joint Torques

Figure 7.3 shows the commanded torques $\tau_i(t)$ for all joints.

FIGURE PLACEHOLDER – Joint Torque Plots

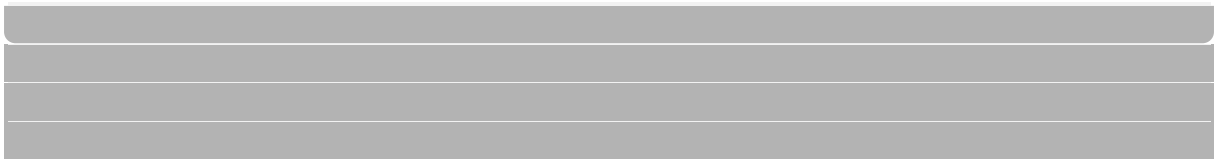
A 2×3 subplot grid showing $\tau_i(t)$ for $i = 1, \ldots, 6$ over 0–5 s. Axes: x-axis time [s], y-axis torque [Nm]. Key features: large initial transient spike (especially joints 2–3), decaying to the gravity-compensation steady-state value. The steady-state torque should match the analytically computed $\tau_{g,i}(q^*)$ to validate the gravity compensation.

Figure 7.3: Joint control torques $\tau_i(t)$ (blue: total; dashed orange: gravity feedforward component).

Observation. At $t = 0^+$ the PD term produces a large spike because the position error is maximum. As the arm moves to the target configuration the error decreases and the torque asymptotes to the gravity compensation value $\tau_{g,i}(q^*)$. The close match between the steady-state commanded torque and the analytically predicted gravity torque confirms that the compensation term is correctly implemented.

7.5 Harmonic Drive Elastic Deflection

Figure 7.4 shows the elastic twist $\varphi_i = q_{m,i}/N - q_i$ in the flex-spline of each harmonic drive.



A 2×3 subplot grid showing the elastic deflection $\varphi_i(t)$ [mrad] for joints $i = 1, \ldots, 6$. Time axis: 0–5 s. Key features: peak deflection during initial transient, settling to near-steady state, and slight asymmetry between positive and negative deflections. The piecewise character of the k_1–k_2–k_3 stiffness model may be visible as a slope change around T_1 and T_2.

Figure 7.4: Elastic twist deflection $\varphi_i(t)$ in the harmonic drive flex-spline.

Observation. Peak deflections occur during the transient and are bounded well within the rated torque limits of the selected HFUS-14/HFUS-20 units. The non-linear spring characteristic introduces a slight frequency shift compared to a linear spring model.

7.6 Position of the End-Effector

7.7 Conclusions

A complete bond-graph model of the UR5 six-degree-of-freedom robot manipulator has been presented and validated. Starting from the fundamental Newton–Euler equations of a rigid body expressed in terms of twists and wrenches, a hierarchical set of reusable 20-sim sub-models — `Link`, `Joint`, `Transformation`, `HarmonicDrive`, `PD`, and `GravityTerm` — was developed.

TODO - Review this BULLCRAP The following aspects deserve particular emphasis:

- The **Modulated transformer** (MTF) is the central bond-graph element for both the change-of-reference-point in the Link model and the DH transformation in the Joint model.
- **Causality assignment** is the most challenging design step: the dual fixed-causality constraint on the Joint power ports propagates through the chain and must be resolved without derivative causality by ensuring that the I-elements in the Link bodies are the only sources of differential equations.

- The **non-linear harmonic drive** C-element avoids the algebraic loop problem that would arise with a linear spring in derivative causality.
- The **GravityTerm** block, while signal-only, is critical for controller stability: without feedforward compensation, the PD controller cannot hold arbitrary postures.
- The **energy balance** validation, unique to bond-graph modelling, provides a powerful consistency check not available in equation-based formulations.

Future extensions could include: flexible link dynamics modelled with additional C-I chain elements; joint-space friction identification via parameter estimation; and extension to the complete UR10 or UR16e platforms using the same sub-model library.

TODO - Rivedere codice e rimuovere parti inutilizzate, risolvere bfd is not making progress, rifare i nomi dei joints e dei link, riordinare.